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<https://leetcode.com/problems/longest-winning-streak/>

Matches table:

player_id	match_day	result
1	2022-01-17	Win
1	2022-01-18	Win
1	2022-01-25	Win
1	2022-01-31	Draw
1	2022-02-08	Win
2	2022-02-06	Lose
2	2022-02-08	Lose
3	2022-03-30	Win

step 1

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player_id	match_day	result	grp_num
1	2022-01-17	Win	0
1	2022-01-18	Win	0
1	2022-01-25	Win	0
1	2022-01-31	Draw	1
1	2022-02-08	Win	1
2	2022-02-06	Lose	1
2	2022-02-08	Lose	2
3	2022-03-30	Win	0

```
select *,
sum(result!='Win') over (partition by player_id order by match_day) as
grp_num
from matches
```

step 2

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player_id	grp_num	streak
1	0	3
1	1	1
3	0	1

```
with grp_cte as (select *,
sum(result!='Win') over (partition by player_id order by match_day) as
grp_num
from matches)
select player_id, grp_num,
```

```
count(1) as streak from grp_cte where result = 'Win'
group by player_id, grp_num;
```

step 3

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player_id	streak	rnum
1	3	1
1	1	2
3	1	1

```
with grp_cte as (select *,
sum(result!='Win') over (partition by player_id order by match_day) as
grp_num
from matches),
window_cte as (select player_id,
count(1) as streak,
row_number() over (partition by player_id order by count(1) desc) as rnum
from grp_cte where result = 'Win'
group by player_id, grp_num)
```

step 4

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player_id	longest_streak
1	3
2	0
3	1

```
with grp_cte as (select *,
sum(result!='Win') over (partition by player_id order by match_day) as
grp_num
from matches),
window_cte as (select player_id,
count(1) as streak,
row_number() over (partition by player_id order by count(1) desc) as rnum
from grp_cte where result = 'Win'
group by player_id, grp_num)
select m.player_id,
case when w.player_id is null
then 0
else w.streak
end as longest_streak
from (select distinct player_id from matches) m
left join (select * from window_cte where rnum = 1) w
on m.player_id = w.player_id
```

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<https://leetcode.com/problems/top-three-wineries/description/>
Wineries table:

id	country	points	winery
103	Australia	84	WhisperingPines
737	Australia	85	GrapesGalore
848	Australia	100	HarmonyHill
222	Hungary	60	MoonlitCellars
116	USA	47	RoyalVines
124	USA	45	EaglesNest
648	India	69	SunsetVines
894	USA	39	RoyalVines
677	USA	9	PacificCrest

step 1

=====

country	winery	total_points
Australia	WhisperingPines	84
Australia	GrapesGalore	85
Australia	HarmonyHill	100
Hungary	MoonlitCellars	60
USA	RoyalVines	86
USA	EaglesNest	45
India	SunsetVines	69
USA	PacificCrest	9

select country, winery, sum(points) as total_points
from wineries
group by country, winery

step 2

=====

country	winery	rnk
Australia	HarmonyHill (100)	1
Australia	GrapesGalore (85)	2
Australia	WhisperingPines (84)	3
Hungary	MoonlitCellars (60)	1
India	SunsetVines (69)	1
USA	RoyalVines (86)	1
USA	EaglesNest (45)	2
USA	PacificCrest (9)	3

```

with points_agg as (select country, winery, sum(points) as total_points
from wineries
group by country, winery)
select country,
concat(winery, " (", total_points, ")") as winery,
rank() over (partition by country order by total_points desc, winery) as rnk
from points_agg

```

step 3

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country	top_winery	second_winery	third_winery	
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Australia	HarmonyHill (100)	GrapesGalore (85)	WhisperingPines (84)	
Hungary	MoonlitCellars (60)	No second winery	No third winery	
India	SunsetVines (69)	No second winery	No third winery	
USA	RoyalVines (86)	EaglesNest (45)	PacificCrest (9)	

```

with points_agg as (select country, winery, sum(points) as total_points
from wineries
group by country, winery),
cte_ranked as (select country,
concat(winery, " (", total_points, ")") as winery,
rank() over (partition by country order by total_points desc, winery) as rnk
from points_agg)
select country,
max(case when rnk = 1 then winery end) as top_winery,
coalesce(max(case when rnk = 2 then winery else null end),'No second
winery')
as second_winery,
coalesce(max(case when rnk = 3 then winery else null end),'No third winery')
as third_winery,
from cte_ranked
group by country;

```